

Backend Engineer Assessment

Your task is to implement a lightweight article service with the following API endpoints:

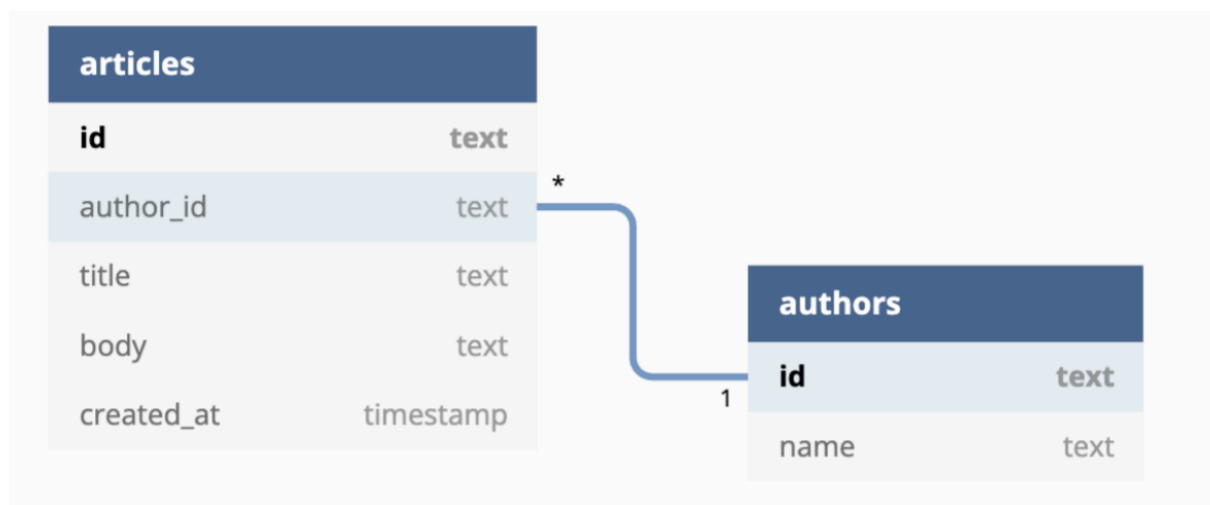
Required Endpoints

1. **POST /articles**
 - Create a new article.
2. **GET /articles**
 - Retrieve a list of articles, sorted by newest first.
 - Support the following optional query parameters:
 - **query**: Search for keywords in the article title and body.
 - **author**: Filter articles by the author's name.

API Protocol

You are free to use **REST**, **gRPC**, or **GraphQL**. Choose the one that best fits your design preferences.

Table Design



Considerations

1. **Scalability.** Design your solution to handle large volumes of articles efficiently.
2. **Concurrency.** Ensure the service performs reliably under high traffic and concurrent access.
3. **Search & Filtering.** Implement fast and accurate keyword search and author-based filtering, considering performance at scale.
4. **Extensibility & Maintainability.** Structure your code and data model to support future features (e.g., categories, comments) and ease of maintenance.

Technical Requirements

- **Use Golang** as the primary language for your solution.
- You may integrate **any widely-used database or search engine** of your choice.
- Include **unit tests and/or integration tests** to validate your implementation.
- (Optional) Use **Docker Compose** to simplify deployment and setup.
- **Do not use any ORM libraries** (e.g., GORM). Please interact with the database using **standard SQL queries**.

Note on AI Usage

We recognize that AI tools (such as ChatGPT, Gemini, Claude, Copilot, etc.) are increasingly used in technical work and exploration. Therefore, we **allow the use of AI to help complete this assessment**, with the following important conditions:

- **Transparency is required:** Please clearly indicate which parts of your work were assisted by AI. For example, whether AI was used for idea exploration, initial code generation, documentation writing, or other tasks.
- **You are fully responsible:** You remain accountable for the quality and understanding of the solution you submit. We will evaluate not just the final result, but also your grasp of the approach you used.
- **We'll know:** Our team has also tested this assessment using various AI tools. So we'll be able to tell if you simply copied AI output without meaningful thought or adaptation.

The main goal of this assessment is to understand your thinking and problem-solving approach. AI can assist, but **it cannot replace your own critical thinking and understanding**.

Timeline & Expectations

You have **3 days** to complete this assessment. While AI tools might help you finish faster, our priority is **quality over speed**. Use the time wisely to refine your approach, validate your results, and ensure your submission reflects your own thinking and understanding.

The goal of this assessment is to understand how you think and solve problems, not just how fast you can generate output.